EEL-5813 Neural Networks FALL 2026

INDIVIDUAL PROJECT I **Perceptron and Adaline** **Due Date: 10/20/2026**

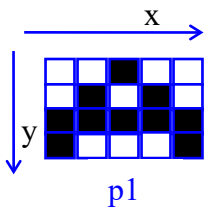
Problem Description and Training Set

In this project you are asked to use the Perceptron and Adaline training algorithms to train a network (a single processing element) for indicating, with an output of +1, when a 4 (rows) by 5 (columns) array of "pixels" that are "ON" (+1) or "OFF" (-1) represents a specific upper-case vowel (for example "A"), from among instances that represent all 5 upper-case vowels.

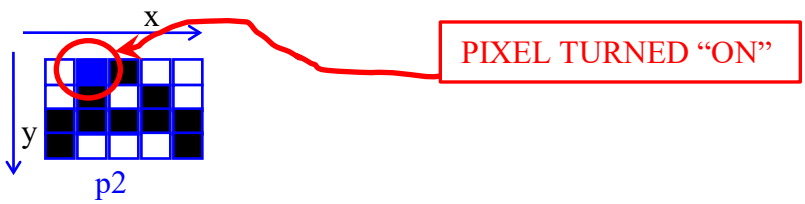
The 20 values (+1 or -1) of the pixels in each of the input patterns will be re-arranged as a column vector of 20 values. Below you can see an example of this re-arrangement:

So, for example:

p1 = [-1 -1 1 1 -1 1 1 -1 1 -1 1 -1 -1 1 1 -1 -1 -1 1 1]^T



The TRAINING SET that you will use for this will be composed by 5 "A" patterns, 5 "E" patterns, 5 "I" patterns, , 5 "O" patterns and 5 "U" patterns. Within each group of 5 patterns for each letter, the first one is a "perfect" character and the remaining 4 are created by turning "ON" only one of the pixels that were originally "OFF". For example, from the "Perfect A", which is pattern p1 shown above, a second "A pattern", p2, is generated by only turning "ON" one pixel (from -1 to +1) as shown below:



Expressed as a column vector:

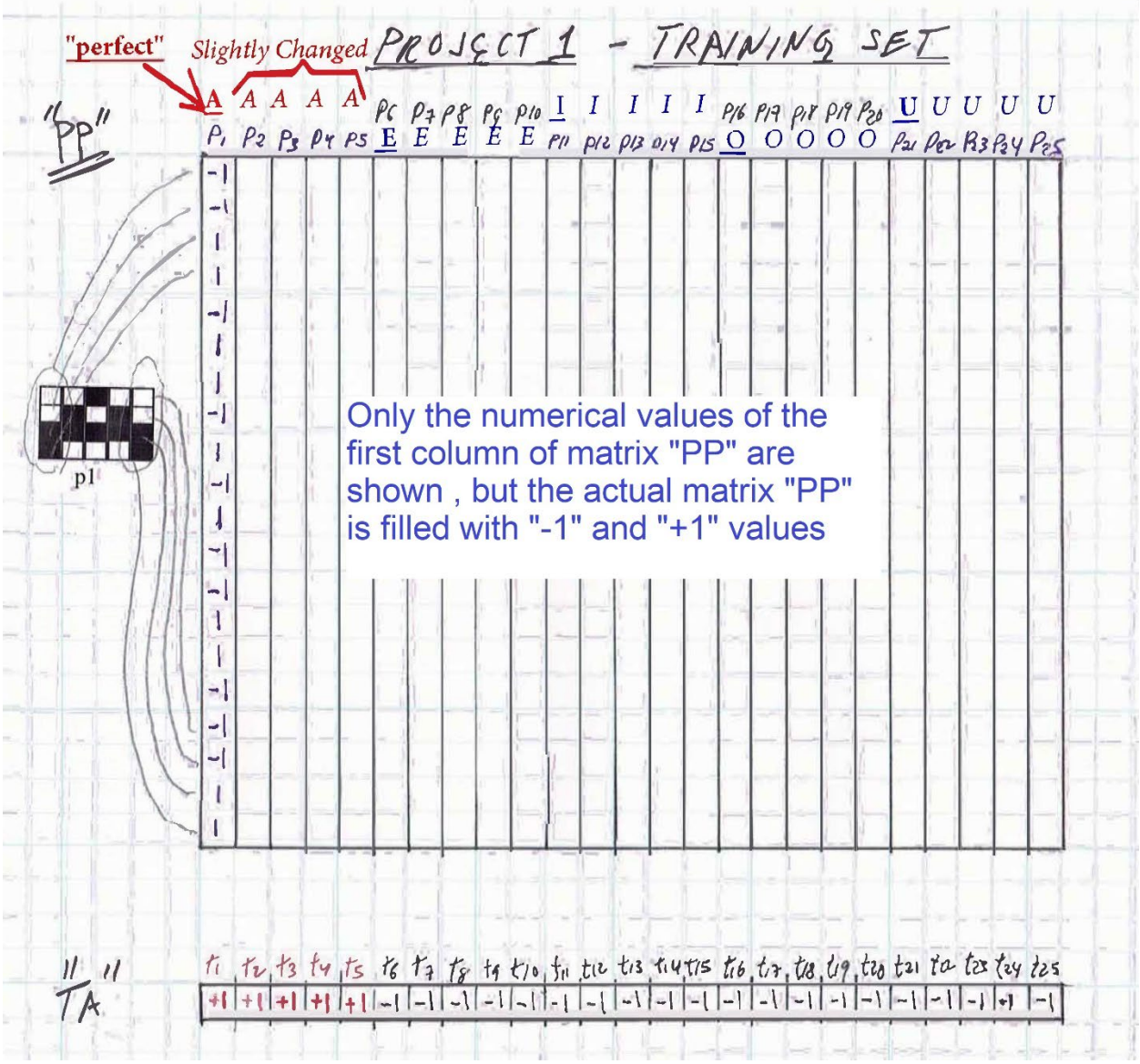
p2 = [-1 -1 1 1 **+1** 1 1 -1 1 -1 1 -1 -1 1 1 -1 -1 -1 1 1]^T

The same approach was followed to:

- Create p3, p4, p5 changing 1 pixel in p1 (This yields 5 "A" TRAINING PATTERNS)
- Create p7, p8, p9, p10 changing 1 pixel in p6 (This yields 5 "E" TRAINING PATTERNS)
- Create p12, p13, p14, p15 changing 1 pixel in p11 (This yields 5 "I" TRAINING PATTERNS)
- Create p17, p18, p19, p20 changing 1 pixel in p16 (This yields 5 "O" TRAINING PATTERNS)
- Create p22, p23, p24, p25 changing 1 pixel in p21 (This yields 5 "U" TRAINING PATTERNS)

All the 25 training patterns are given to you[1] in a 20 (rows) by 25 (columns) matrix "PP" where each column is a training pattern already rearranged as a column. The next page shows how the matrix PP is constituted and it also shows the 1 (row) by 25 (columns) matrix "TA" that provides the targets to be used when the system will be trained to identify input patterns that represent an "A".

[1] All the matrices that "are given to you" are available in the Canvas page for this project as individual Matlab data files (e.g., PP.mat) or CSV text files (e.g., PP.txt). To load a matrix from a file in Matlab use the instruction load, for example load('PP.m')



When you will train the network to be an “A Detector”, you will use the columns of PP as input patterns and the corresponding entries in TA as targets. For example, you will use the first column, p1, at the input of the network and t1 = **+1** for target. Similarly, you will use p2 and t2 = **+1**, p3 and t3 = **+1**, p4 and t4 = **+1**, p5 and t5 = **+1**, p6 and t6 = **-1**, etc.

You are also given a 1 (row) by (25) columns matrix “TE” that provides you with the targets needed to train the same network, with the same patterns from PP as input, so that the network will set its output activation to +1 when the pattern presented is an “E”. (I.e., to train the network as an “E Detector”).

The figure below shows how the “perfect” patterns (p1, p6, p11, p16, and p21) were modified (by turning “ON” 1 pixel in each case) to create 4 additional samples of each vowel, so that there are 5 patterns for each vowel in matrix PP. (This is for informational purposes only, since the complete matrix PP is already created for you).

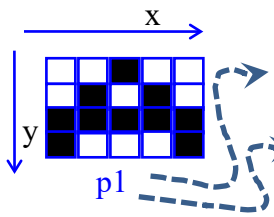
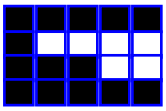
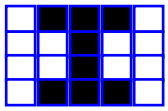
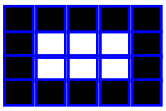
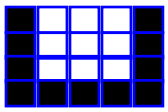
To create This training pattern	Turn ON this pixel (x,y)
	p2 (2,1) p3 (3,2) p4 (3,4) p5 (4,4)
 p6	p7 (2,2) p8 (4,2) p9 (5,3) p10 (4,3)
 p11	p12 (1,1) p13 (4,2) p14 (1,4) p15 (4,3)
 p16	p17 (2,2) p18 (4,2) p19 (2,3) p20 (4,3)
 p21	p22 (2,1) p23 (4,1) p24 (2,3) p25 (4,3)

FIGURE 1: Creation of the TRAINING (INPUT) PATTERNS
(YOU WILL RECEIVE THE 20-by-25 MATRIX WITH THESE PATTERNS, in matrix PP)

PART I: PERCEPTRON:

I.1 "A DETECTOR" – Single PE to indicate the presence of an "A" pattern at the input

You will train and test a PERCEPTRON with 20 weights, a bias and a single output activation. This will be an "A DETECTOR" Therefore the 5 targets for the training patterns that represent variations of "A" will be +1. The remaining targets (those for patterns that represent variations of "E", "I" "O" or "U") will be -1: That is:

t1 = t2 = t3 = t4 = t5 = +1

t6 = t7 = t8 = t9 = t10 = t11 = t12 = t13 = t14 = t15 = t16 = t17 = t18 = t19 = t20 =t21=t22 = t23 = t24 = t25 = -1

These target values are already encoded in the matrix TA (given to you).

Train the perceptron, KEEPING TRACK OF HOW MANY OF THE 25 TRAINING PATTERNS PROMPTED A CHANGE OF WEIGHTS ($e \neq 0$) in each epoch of training – Include a PLOT of these numbers, from epoch 1 to the last epoch you trained. SHOW A TABLE (in column format) of your final weights and bias. – Since the target values are either +1 or -1, you must use the "*bardlims*" nonlinear activation function.

- I.1 Testing your Perceptron "A Detector"

You will TEST your trained perceptron SEPARATELY WITH EACH ONE of the 3 progressively more difficult test sets (Test Set 1, Test Set 2, Test Set 3) which are given to you as matrices TST1, TST2, TST3. Each of these matrices has the same structure of 20 (rows) by 25 (columns) as PP, and each of the columns of these matrices is a different test pattern. Just as for the matrix PP, the first 5 columns are "A" patterns, columns 6 to 10 are "E patterns", etc. Accordingly, the same TA matrix can be consulted to know what the correct output from the network will be for any column of the test matrices (TST1, TST2, TST3).

In EACH CASE (i.e., when you test with EACH of the test matrices), LIST THE PATTERNS CLASSIFIED INCORRECTLY by the trained perceptron (Each pattern can be named as "pSnn", where S is the Test Set number and nn is the 2-digit column number. For example, the pattern in column 8 of test set 2 will be p208). -In each case, you must also indicate the "Hit Ratio" (accuracy) obtained:

HR = Hit Ratio = # correct classifications / # attempted classifications = # correct classifications / 25

I.2 "E DETECTOR" – Single PE to indicate the presence of an "E" pattern at the input

To create a perceptron "E Detector" you will use the same architecture as for I.1 and the same set of 25 training patterns (p1, …p25).

THE ONLY CHANGE NEEDED is a change of the target values:

t1 = t2 = t3 = t4 = t5 = -1 (because p1, …, p5 are A s)

t6 = t7 = t8 = t9 = t10 = +1(because p6, ..p10 are the "E patterns" this network should detect)

t11 = t12 = t13 = t14 = t15 = t16 = t17 = t18 = t19 = t20 =t21=t22 = t23 = t24 = t25 = -1 (I,O,U)

This change of target values can be accomplished simply by using the 1 (row) by 25 (columns) matrix "TE" given to you, instead of the matrix TA that you used for the "A Detector".

You will test this "E DETECTOR" as you did for the "A DETECTOR".
You must report the same set of results as for the "A detector".

I.3 COMMENTS: Write comments and observations about the training and testing of the "A DETECTOR" and the "E DETECTOR". Comment on any differences between the two. Provide overall comments on the complete perceptron experimentation.

PART II: ADALINE:

II.1 "A DETECTOR" – Single PE to indicate the presence of an "A" pattern at the input

You will now train the same architecture (1 PE, 20 weights, 1 bias), using the Adaline (Widrow-Hoff) training algorithm. Accordingly, during training, a LINEAR (*purelin*) activation function will be used (a = n). In this case, however, you must define the value for the parameter alpha to be able to train the network. You are asked to TENTATIVELY train using several values of alpha and observe the difference in the progression of training

- In this case, instead of the number of training patterns that prompt changing of the weights (which you monitored and displayed for the perceptron), YOU WILL CALCULATE THE MEAN SQUARED ERROR in each epoch of training and save those values to be able to plot the graph of MSE vs. epoch of training for all the training epochs you allowed (i.e., the "Learning Curve").
- You must ALSO plot the evolution of the values of the BIAS and 4 of the 20 weights in the network at the end of each epoch.(Pick one weight corresponding to each of the 4 "rows" in each 4 x 5 image. For example, plot the values of weights w1, w2, w3 and w4)

THIS MEANS YOU MUST INCLUDE IN YOUR REPORT the plot of MSE vs. epoch (the "Learning Curve") and the plots of the 4 weights and the bias for at least 3 values of alpha (which must be stated for each plot, i.e., α = ..):

a) One that ended up being "too small"
b) One that ended up being "too large"
c) The one that you believe gave you the "best trained weights", converging at a reasonable speed.

Then you will ONLY USE THE "BEST TRAINING WEIGHTS" for testing the Adaline detector with the 3 test sets, as you did for Perceptron – TO OBTAIN A DEFINITIVE ANSWER, APPLY A THRESHOLD AT 0 TO THE OUTPUT OF ADALINE DURING TESTING (if n >= 0 , result = +1; otherwise, result = -1). This means you will use *bardlims( )*, BUT ONLY DURING TESTING.

As you did when testing the perceptron, FOR EACH TEST SET list all patterns classified incorrectly and calculate and display the Hit Ratio (the accuracy).

II.2 "E DETECTOR" – Single PE to indicate the presence of an "E" pattern at the input

To create an Adaline "E Detector" you will use the same architecture as for II.1 and the same set of 25 training patterns (p1, …p25).

THE ONLY CHANGE NEEDED is a change of the target values:

t1 = t2 = t3 = t4 = t5 = -1 (because p1, …, p5 are A s)

t6 = t7 = t8 = t9 = t10 = +1(because p6, ..p10 are the "E patterns" this network should detect)

t11 = t12 = t13 = t14 = t15 = t16 = t17 = t18 = t19 = t20 =t21=t22 = t23 = t24 = t25 = -1 (I,O,U)

This change of target values can be accomplished simply by using the 1 (row) by 25 (columns) matrix "TE" given to you, instead of the matrix TA that you used for the "A Detector".

You will test this "E DETECTOR" as you did for the "A DETECTOR".
You must report the same set of results as for the "A detector".

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